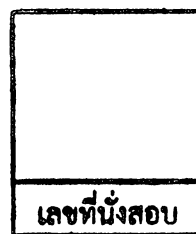


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2017

Subject: 392133 Physics III

Section: 15-16

Date: 30 November 2017

Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 7 pages (including this page), 6 problems and a total score of 60 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.

1.1 [5 points] A 180 cm man wishes to see a full length image of himself in a plane mirror. Find the required minimum length mirror.

1.2 [5 points] A concave mirror has a focal length of 20 cm. What is the magnification if the object's distance is 100 cm? Draw a ray diagram to indicate the image position.

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2.1 [5 points] An object 50 cm high is placed 1.0 m in front of a converging lens whose focal length is 1.5 m. Determine the image height (in cm). Draw a ray diagram to locate the image position.

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2.2 [5 points] A plane convex lens is made of glass ($n = 1.5$) with one flat surface and the other having a radius of 20 cm. What is the focal length (in cm) of the lens?

3.1 [5 points] A converging lens of focal length 20 cm is placed in contact with a diverging lens of focal length 30 cm. Find the focal length of this combination. Describe an experimental approach to determine the focal length of this combination.

3.2 [5 points] A far-sighted student has a near point of 50 cm. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm).

4.1 [5 points] A 10 cm focal length converging lens is used to magnify small newspaper print 2.0 mm high. Calculate the height of the image (in mm) for the maximum magnification for a normal eye.

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4.2 [5 points] In a Young's double-slit experiment, the slit separation is doubled. To Find the screen-to-slit distance L that must be changed to maintain the same fringe spacing on the screen.

5.1 [5 points] Monochromatic light, at normal incidence, strikes a thin film in air. If λ denotes the wavelength in the film, what is the thinnest film in which the reflected light will be at maximum?

5.2 [5 points] Light from a helium-neon laser ($\lambda = 632.8$ nm) is incident upon a 0.200 mm wide slit. Find the total width of the central maximum 2.00 m from the slit.

6.1 [5 points] Monochromatic light is normally incident on a diffraction grating that is 1 cm wide and has 10,000 slits. The first order line is deviated at a 30° angle. What is the wavelength of the incident light?

6.2 [5 points] In a stack of three polarizing sheets the first and third are crossed while the middle one has its axis at 45° to the axes of the other two. Find the fraction of the intensity of an incident unpolarized beam of light that is transmitted by the stack.